

Processes



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Process Concept

- An operating system executes a variety of programs:
 - Batch system – jobs
 - Time-shared systems – user programs or tasks
- The terms job and process are almost interchangeable.
- **Process** – a program in execution; process execution must progress in sequential fashion
- **Program**- a piece of code which may be a single line or millions of lines
- A process includes:
 - program counter
 - stack
 - data section

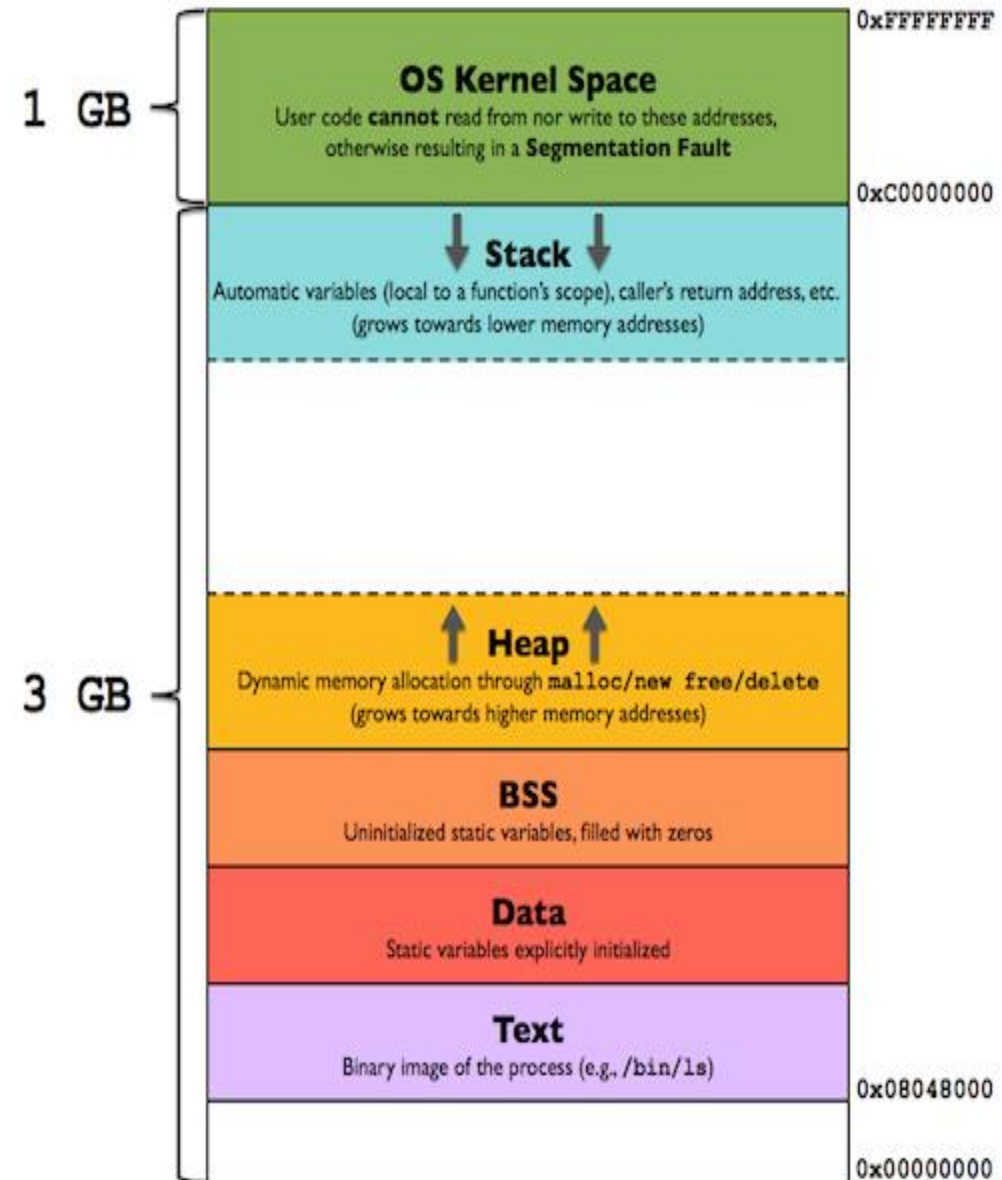
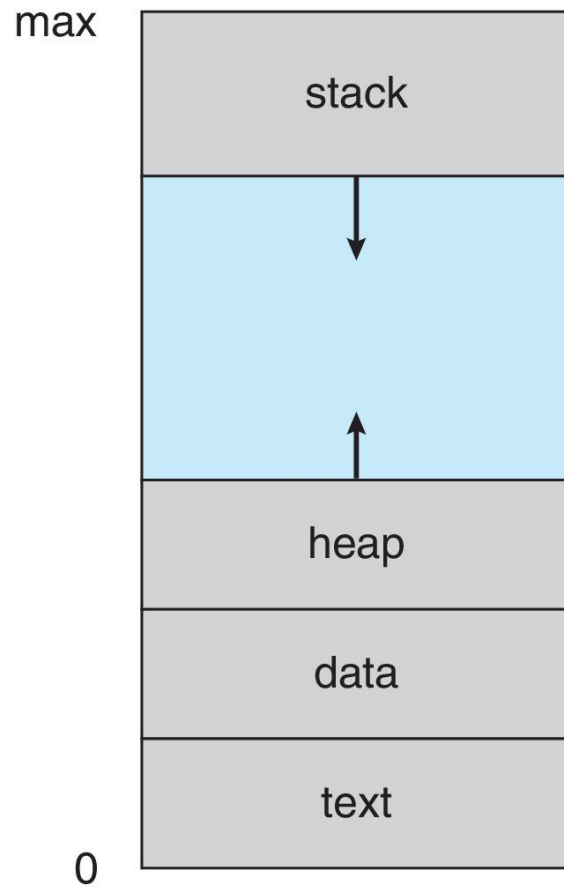
Process Concept

- An operating system executes a variety of programs that run as a process.
- **Process** – a program in execution; process execution must progress in sequential fashion. No parallel execution of instructions of a single process
- Multiple parts
 - The program code, also called **text section**
 - Current activity including **program counter**, processor registers
 - **Stack** containing temporary data
 - Function parameters, return addresses, local variables
 - **Data section** containing global variables
 - **Heap** containing memory dynamically allocated during run time

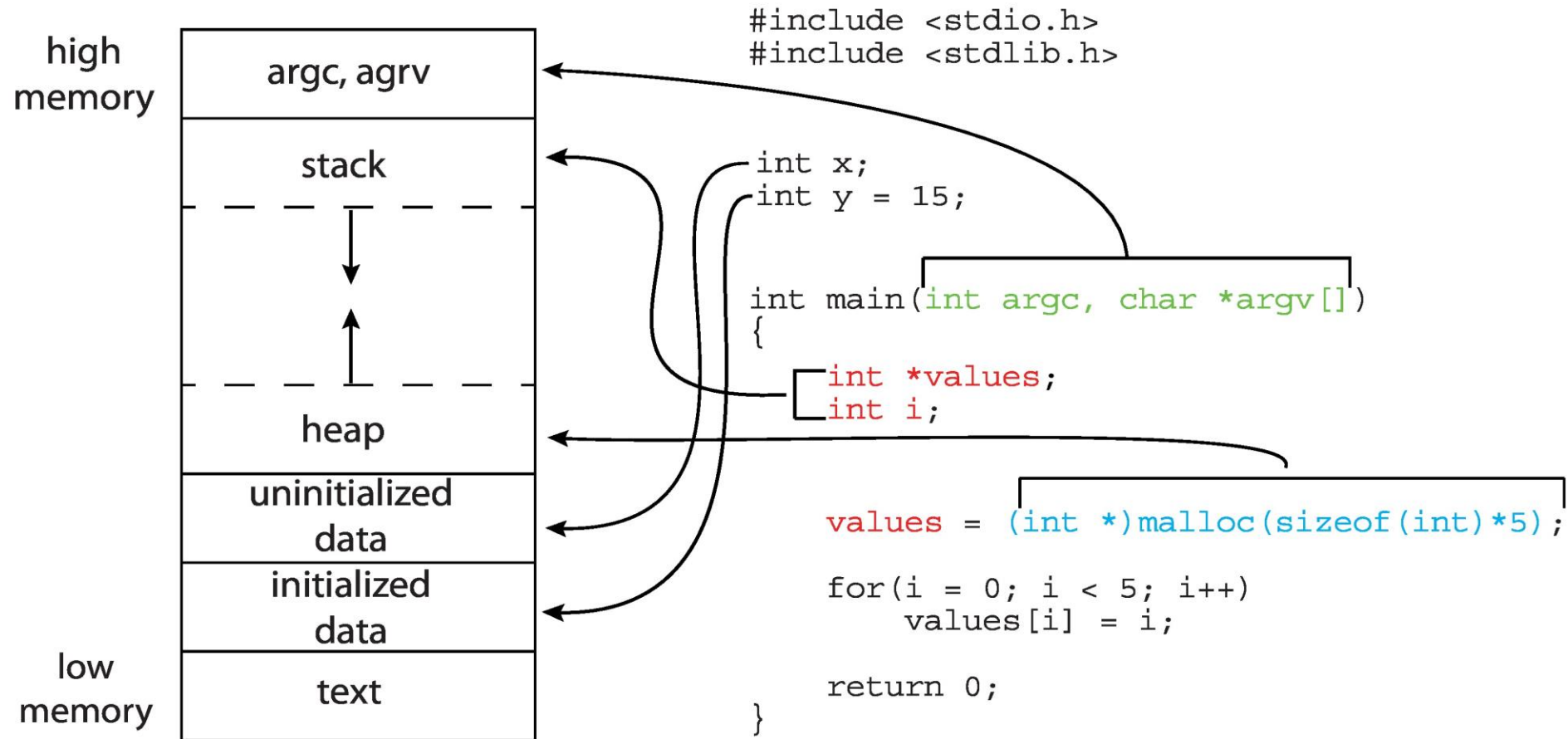
Process Concept (Cont.)

- Program is **passive** entity stored on disk (**executable file**); process is **active**
 - Program becomes process when an executable file is loaded into memory
- Execution of program started via GUI mouse clicks, command line entry of its name, etc.
- One program can be several processes
 - Consider multiple users executing the same program
 - Compiler
 - Text editor

Process in Memory



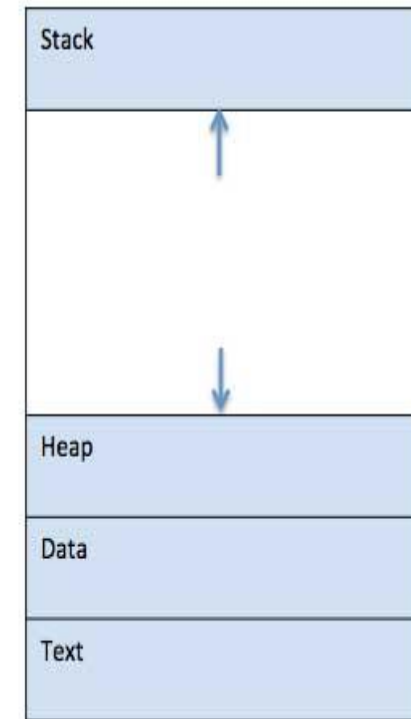
Memory Layout of a C Program



Layout of a process inside main memory

- When a program is loaded into the memory and it becomes a process, it can be divided into four sections – stack, heap, text and data.

S.N.	Component & Description
1	Stack The process Stack contains the temporary data such as method/function parameters, return address and local variables.
2	Heap This is dynamically allocated memory to a process during its run time.
3	Text This includes the current activity represented by the value of Program Counter and the contents of the processor's registers.
4	Data This section contains the global and static variables.



Process State

- As a process executes, it changes *state*
 - **new**: The process is being created
 - **running**: Instructions are being executed
 - **waiting**: The process is waiting for some event to occur
 - **ready**: The process is waiting to be assigned to a process
 - **terminated**: The process has finished execution

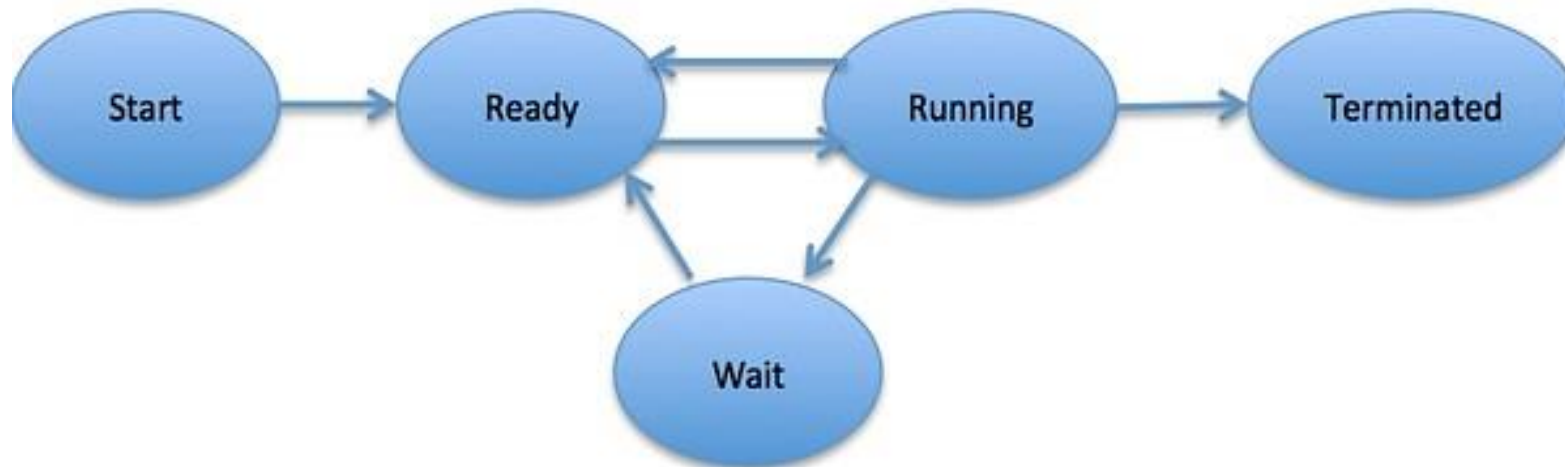
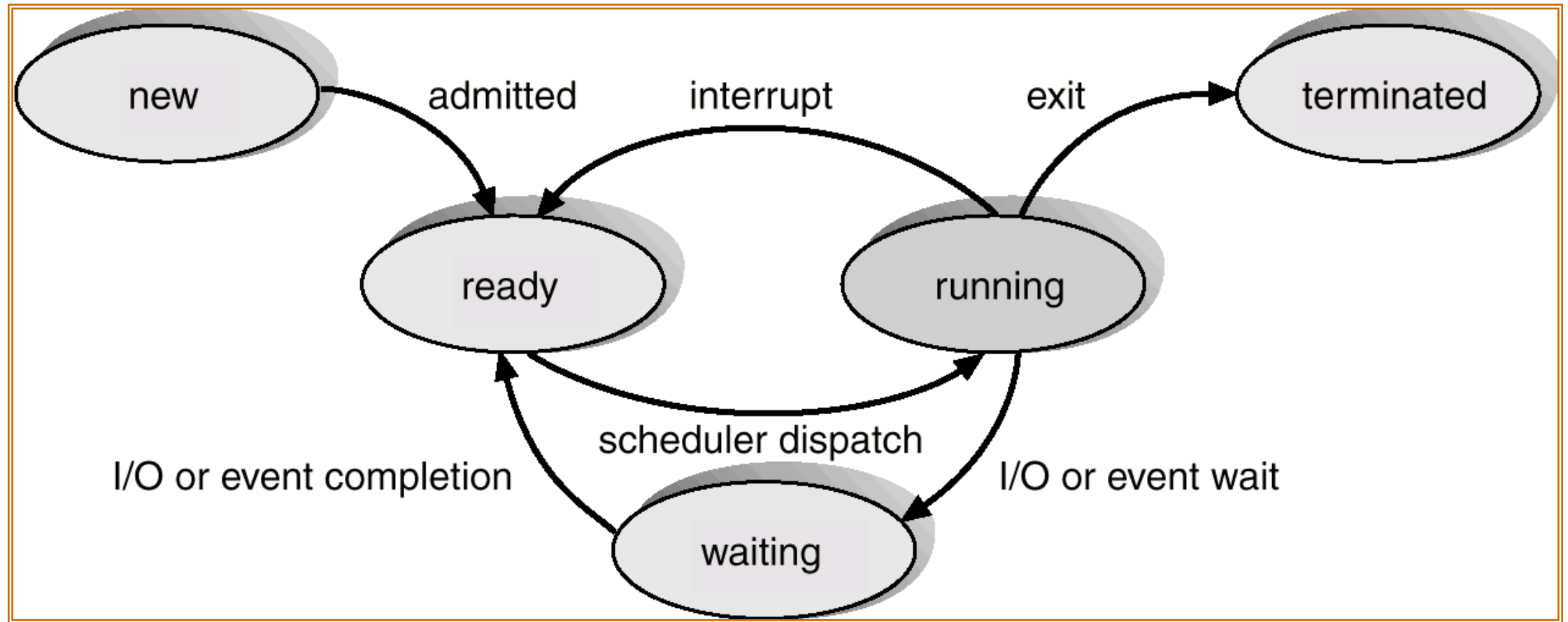


Diagram of Process State



Different time with respect to a process

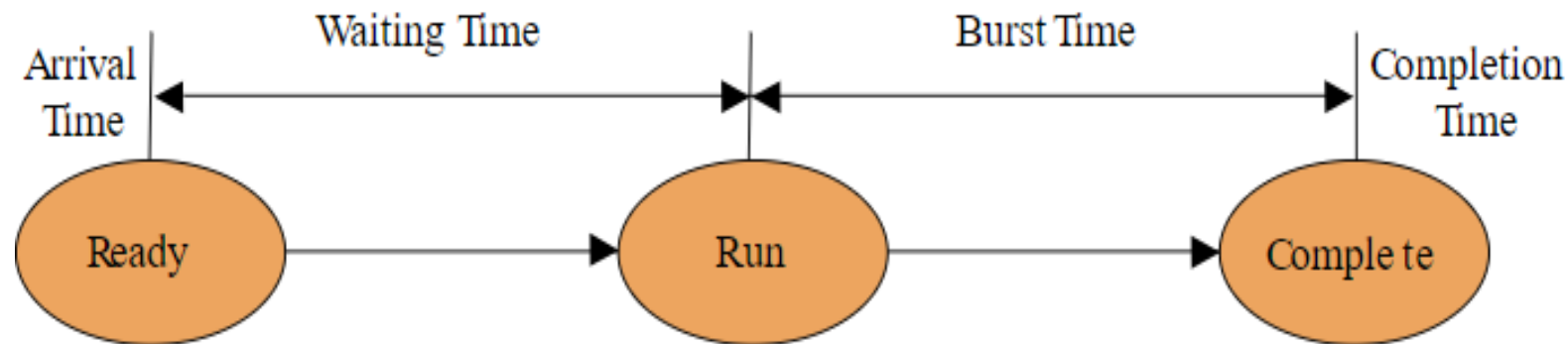
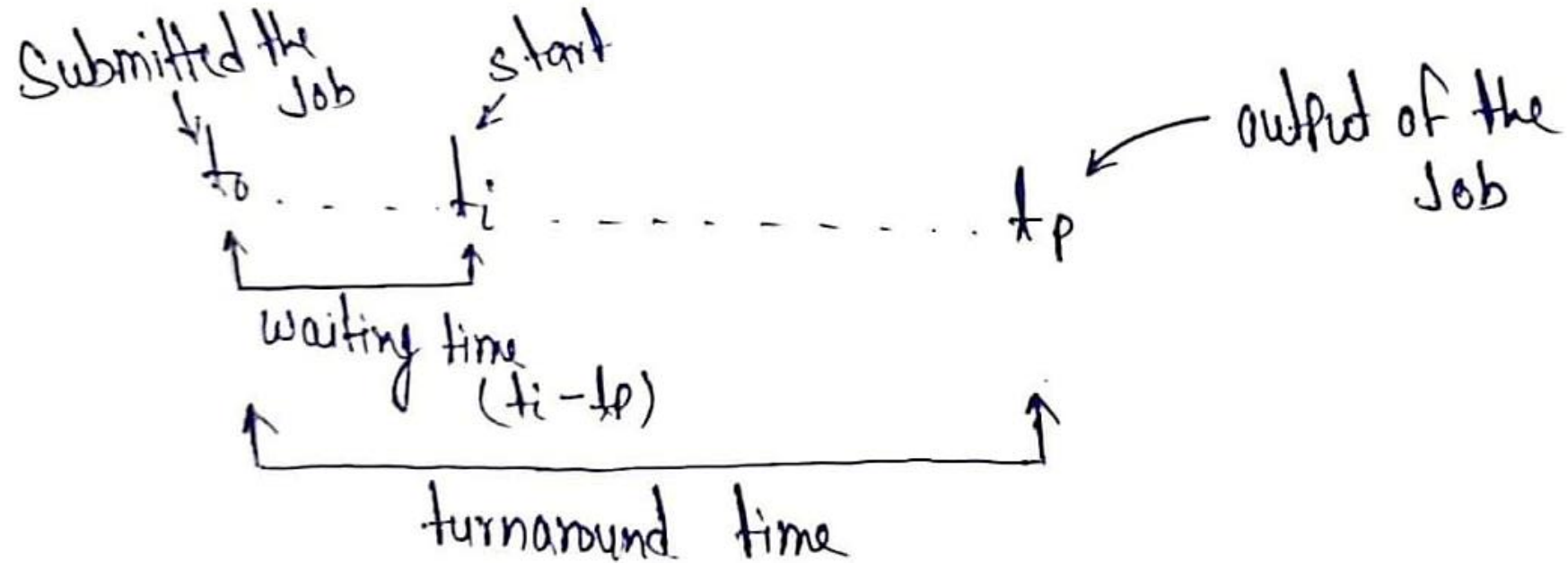
- Arrival Time: Time at which the process arrives in the ready queue.
- Completion Time: Time at which process completes its execution.
- Burst Time: Time required by a process for CPU execution.
- Turn Around Time: Time Difference between completion time and arrival time.

$$\text{Turn Around Time} = \text{Completion Time} - \text{Arrival Time}$$

- Waiting Time(W.T): Time Difference between turn around time and burst time.

$$\text{Waiting Time} = \text{Turn Around Time} - \text{Burst Time}$$

Waiting vs Turnaround time



$$TAT = CT - AT$$

$$CT - AT = WT + BT$$

$$\text{Waiting Time} = TAT - BT$$

FCFS vs SJF

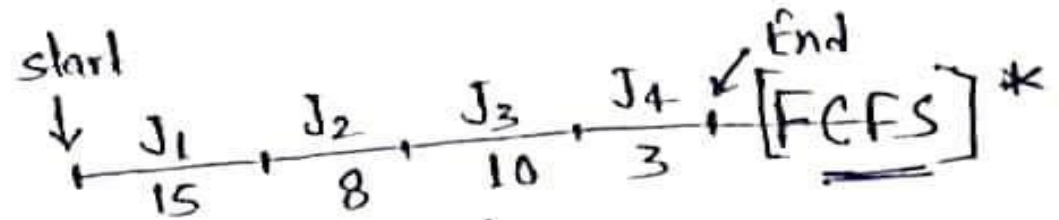
Example: -

First- $J_1 - 15$

Second $J_2 - 8$

$J_3 - 10$

Last $J_4 - 3$
Job Execution time



Waiting time of Job 1, $w_{J_1} = 0$

$$w_{J_2} = 15$$

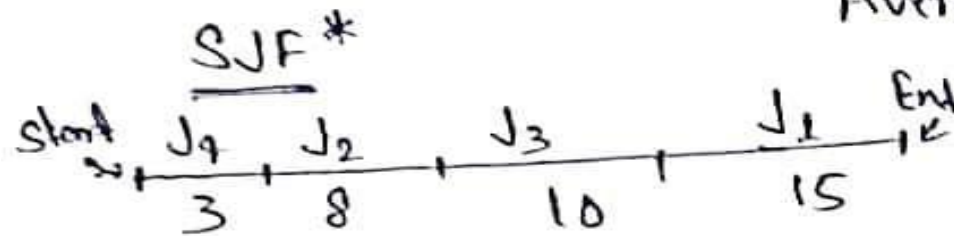
$$w_{J_3} = 23$$

$$w_{J_4} = 33$$

71

Average waiting time $t_{w_A} = \frac{71}{4}$

$$= 17.75$$



$$w_{J_4} = 0$$

$$w_{J_2} = 3$$

$$w_{J_3} = 11$$

$$w_{J_1} = 21$$

35

Average waiting time $t_{w_A} = \frac{35}{4}$

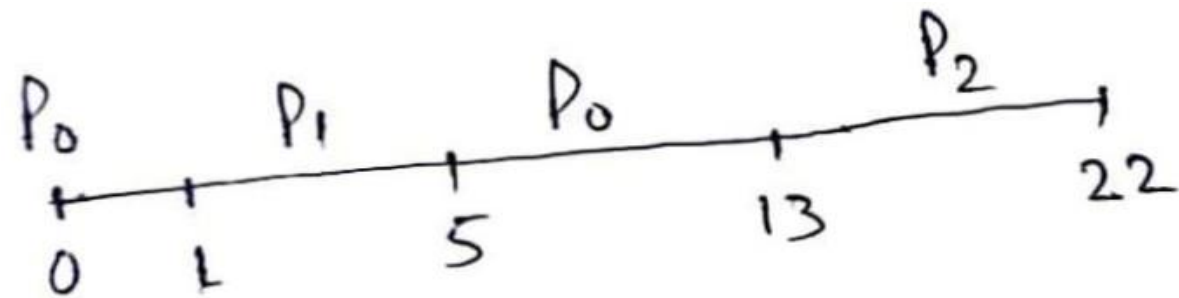
$$= 8.75$$

Consider the following table of arrival time and burst time for three processes P0, P1 and P2. (GATE-CS-2011)

Process	Arrival time	Burst Time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

- (A) 5.0 ms
- (B) 4.33 ms
- (C) 6.33
- (D) 7.33



$$\begin{aligned}w_{P_0} &= 0 + 4 = 4 \\w_{P_1} &= 0 \\w_{P_2} &= 13 - 2 = 11\end{aligned}$$

$$t_{w_A} = \frac{15}{3} = \underline{\underline{5ms}}$$

Consider the following set of processes, with the arrival times and the CPU-burst times given in milliseconds (GATE-CS-2004)

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	3
P4	4	1

What is the average turnaround time for these processes with the preemptive shortest remaining processing time first (SRPT) algorithm ?

- (A) 5.50
- (B) 5.75
- (C) 6.00
- (D) 6.25

An operating system uses the Shortest Remaining Time first (SRTF) process scheduling algorithm. Consider the arrival times and execution times for the following processes: (GATE-CS-2007)

Process	Execution time	Arrival time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

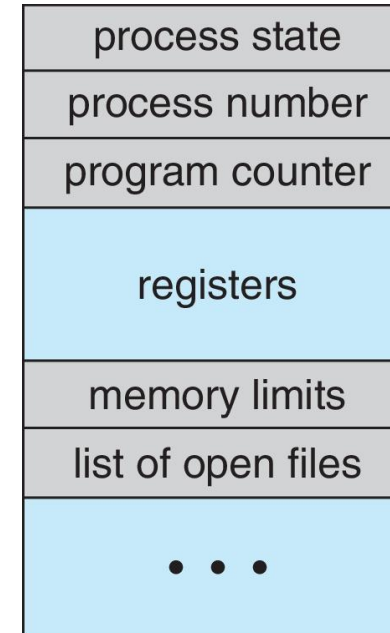
What is the total waiting time for process P2?

- (A) 5
- (B) 15
- (C) 40
- (D) 55

Process Control Block (PCB)

Information associated with each process (also called [task control block](#))

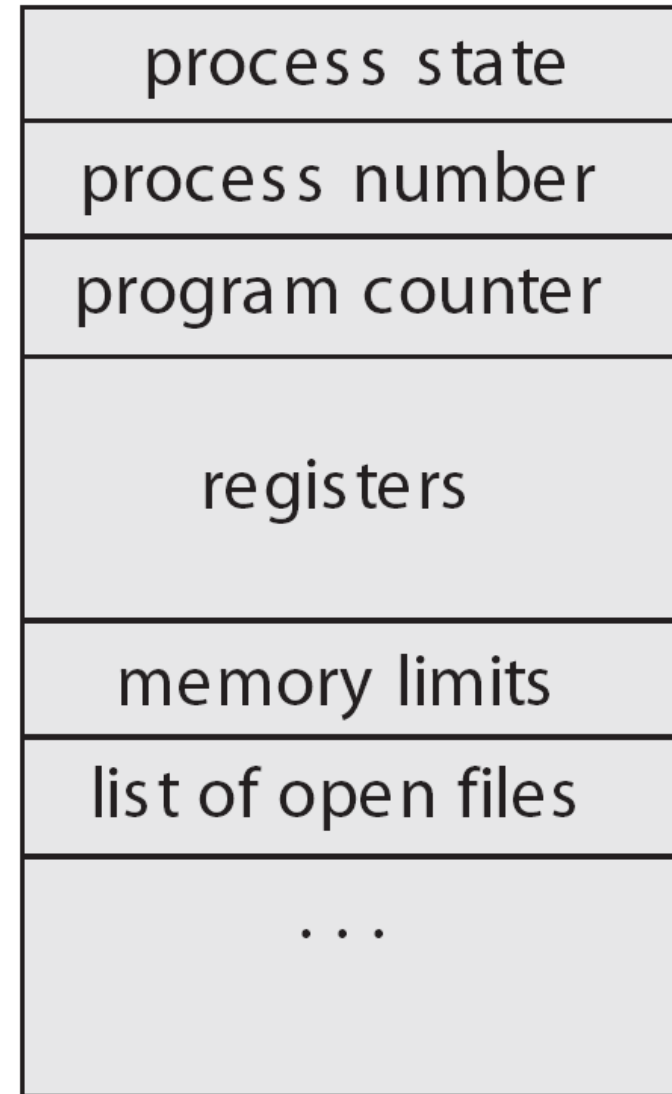
- **Process state** – running, waiting, etc.
- **Program counter** – location of instruction to next execute
- **CPU registers** – contents of all process-centric registers
- **CPU scheduling information**- priorities, scheduling queue pointers
- **Memory-management information** – memory allocated to the process
- **Accounting information** – CPU used, clock time elapsed since start, time limits
- **I/O status information** – I/O devices allocated to process, list of open files



Process Control Block (PCB)

Information associated with each process

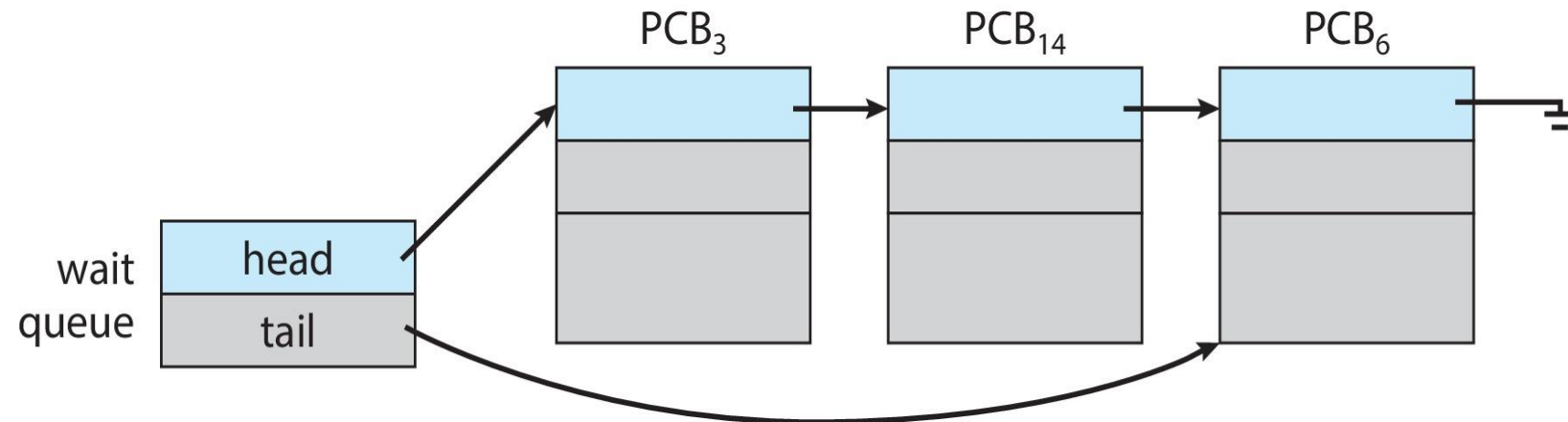
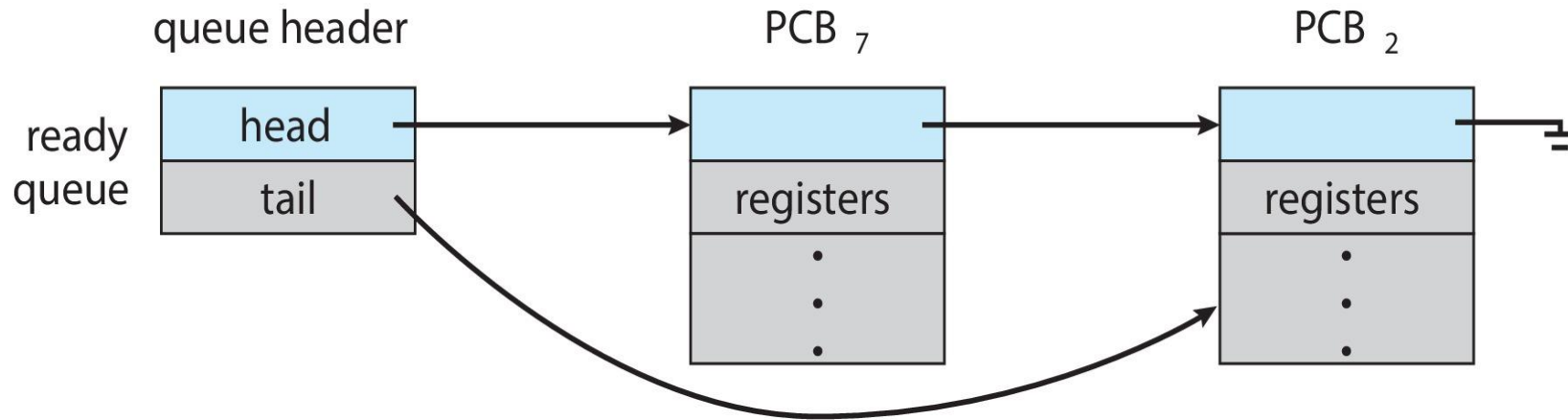
- Process state
- Program counter
- CPU registers
- CPU scheduling information
- Memory-management information
- Accounting information
- I/O status information



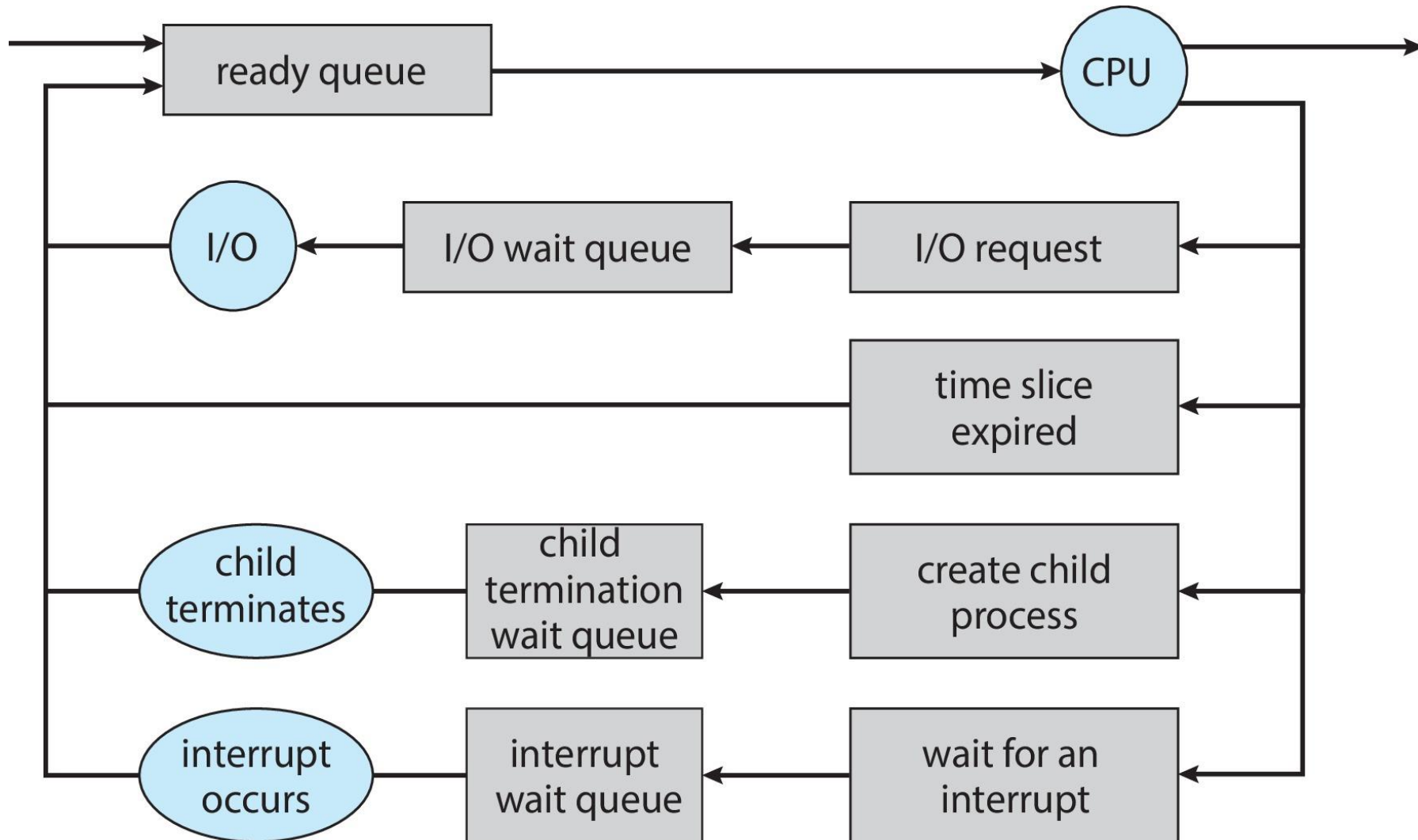
Process Scheduling

- **Process scheduler** selects among available processes for next execution on CPU core
- Goal -- Maximize CPU use, quickly switch processes onto CPU core
- Maintains **scheduling queues** of processes
 - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
 - **Wait queues** – set of processes waiting for an event (i.e., I/O)
 - Processes migrate among the various queues

Ready and Wait Queues

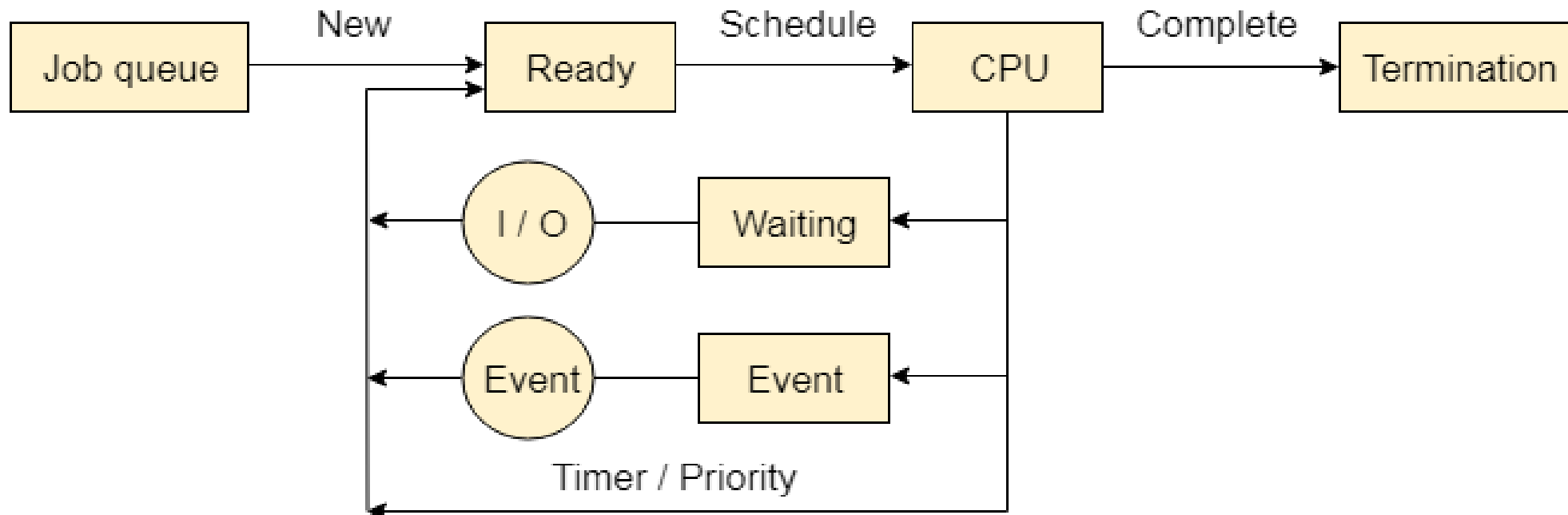


Representation of Process Scheduling

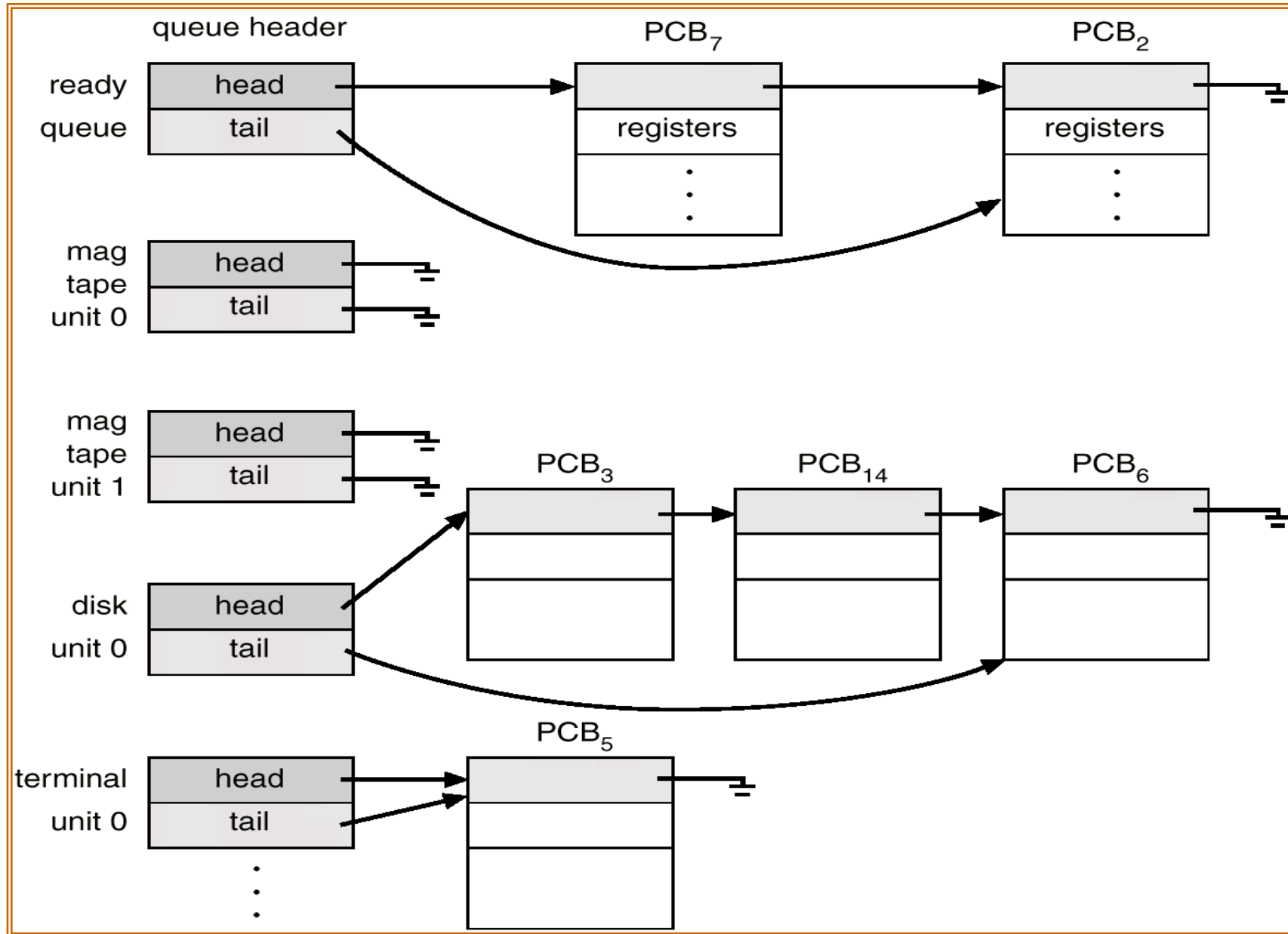


Process Scheduling Queues

- *Job queue* – set of all processes in the system
- *Ready queue* – set of all processes residing in main memory, ready and waiting to execute
- *Device queues* – set of processes waiting for an I/O device
- Process migration between the various queues



Ready Queue And Various I/O Device Queues

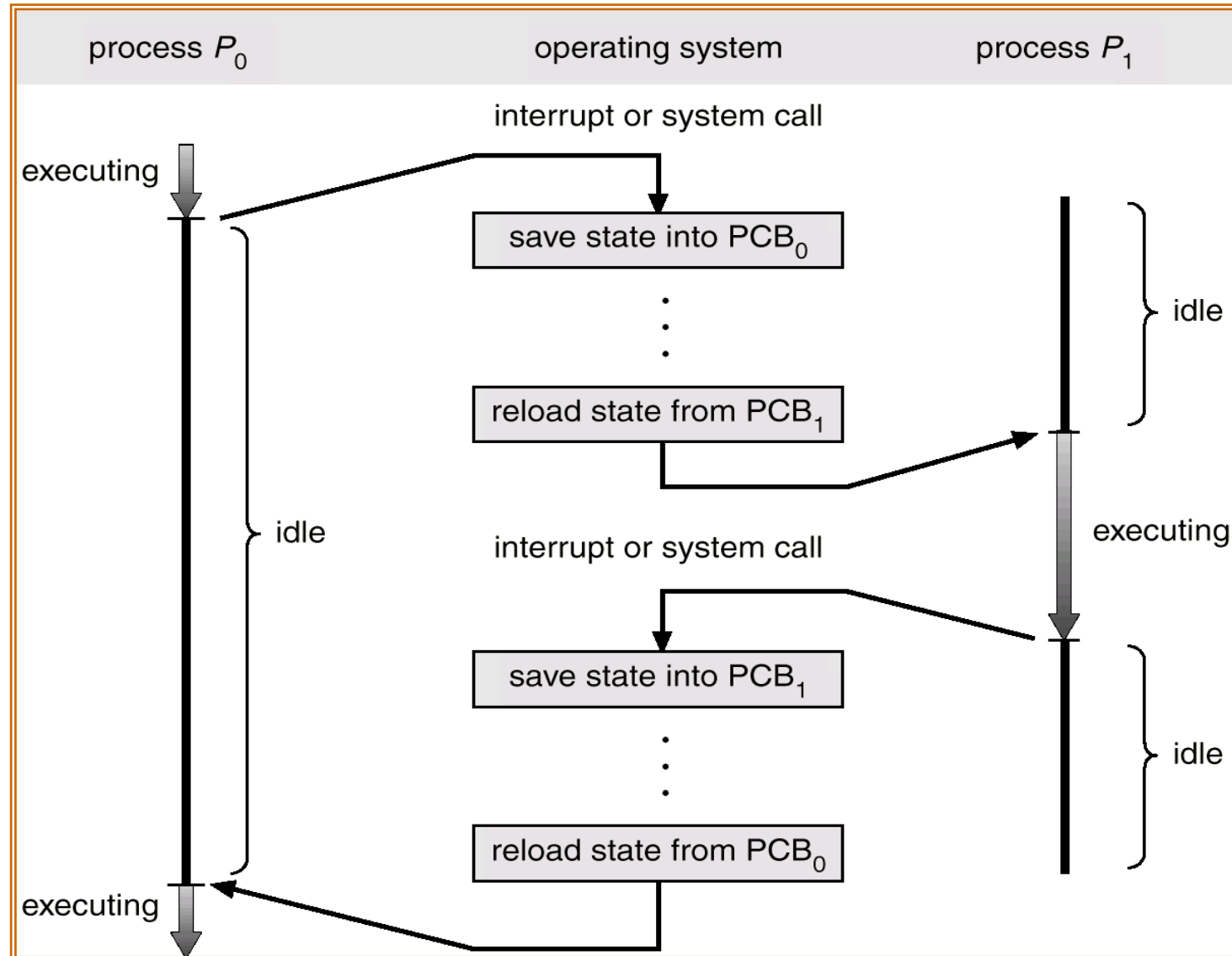


Context Switch

- When CPU switches to another process, the system must **save the state** of the old process and load the **saved state** for the new process via a **context switch**
- **Context** of a process represented in the PCB
- Context-switch time is pure overhead; the system does no useful work while switching
 - The more complex the OS and the PCB → the longer the context switch
- Time dependent on hardware support
 - Some hardware provides multiple sets of registers per CPU → multiple contexts loaded at once

CPU Switch From Process to Process

A **context switch** occurs when the CPU switches from one process to another.



Thanks